MIL-B-37245A
14 May 1976
SUPERSEDING
MIL-B-37245
6 June 1975

MILITARY SPECIFICATION

BUPIVACAINE HYDROCHLORIDE INJECTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Bupivacaine Hydrochloride Injection.

1.2 Classification. The injection covered by this specification shall be the following strengths and sizes (see 6.2):

Strength 1 - 0.25%
Strength 2 - 0.50%
Strength 3 - 0.75%
Size 1 - 50 ml
Size 2 - 30 ml, 5s
Size 3 - 50 ml, 5s

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186 Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals. (see 6.3)

FSC 6505

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STANDARDS

Federal

FED-STD-142 Parenteral Preparations.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

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3. REQUIREMENT

3.1 Material. Shall be Bupivacaine Hydrochloride Injection and shall be in accordance with all requirements of FED-STD-142 and as specified herein:

S2. Classification. Shall be type I, class 1.

S5.2.2 The following is added to this paragraph:

Shall be a sterile solution of Bupivacaine Hydrochloride, 0.25% for strength 1, 0.50% for strength 2 and 0.75% for strength 3, within the designated assay limits for the injection.

Shall be suitable for caudal , epidural or peripheral nerve block.

3.1.1 Assay. The injection shall assay to contain not less than 95.0 percent and not more than 105.0 percent of the required amount of Bupivacaine Hydrochloride, when determined as described in 4.3.1.

3.1.2 Identification. The retention time of the Bupivacaine Hydrochloride shall match that of the reference standard similarly treated (see 4.3.1).

3.1.3 pH. The pH of the injection shall not be less than 4.0 and not more than 6.5 at 25°C. Determination shall be on the undiluted finished product.

3.2 Ingredients.

3.2.1 Bupivacaine Hydrochloride. The Bupivacaine Hydrochloride used in the manufacture of the finished injection shall comply with the following requirements:

3.2.1.1 Melting range. Shall melt between 247° and 262°C with decomposition when determined using the USP Class Ia method without previous drying of the sample.

3.2.1.2 Heavy Metals. Shall not be more than 20 p.p.m. when determined using the U.S.P. Method II.

3.2.1.3 Identification. The injection shall comply with the requirements of 4.3.2.1 and 4.3.2.2.

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3.2.1.4 Assay. Shall assay not less than 98.5 percent and not more than 102.0 percent when determined as specified in 4.3.2.3.

3.2.2 Other ingredients. All other ingredients used in the manufacture of the injection shall comply with S5.1 of FED-STD-142.

3.3 Expiration dating period. The expiration dating period shall be not less than 24 months.

3.3.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with FED-STD-142.

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4.3 Tests.

4.3.1 Bupivacaine Hydrochloride (Identification and Assay).

Gas Chromatographic Parameters: (will vary with instruments)

Gas Chromatograph: F & M Model 700 or equivalent
Column: 6 ft x 6 mm o.d. glass
Column Packing: 3% Hi-Eff-3BP (Neopentyl Glycol Succinate) on 100/120 mesh Gas Chrom Q
Column Temperature: 200°C
Injection Port Temperature: 250°C
Detector: Flame Ionization
Detector Temperature: 250°C
Helium Cylinder Pressure: 30 psig
Air Cylinder Pressure: 15 psig
Hydrogen Cylinder Pressure: 22 psig
Attenuation: x 500

Standard Solution: Dissolve accurately weighed quantities of Bupivacaine Hydrochloride Reference Standard in purified water, and dilute quantitatively with purified water to obtain a Standard Solution having approximately the same concentration as the sample to be analyzed.

Gas Chromatographis Assay: Inject exactly 5.0 µl of the Standard Solution onto the chromatographic column, and observe the peak for Bupivacaine Hydrochloride Reference Standard. Inject exactly 5.0 µl of the sample onto the chromatographic column, and observe the peak for Marcaine Hydrochloride. Calculate the quantity, in mg, of the component by the formula $C(A_U/A_S)$, in which C is the concentration, in mg per ml, of the component in the Standard Solution, and A_U and A_S are the chromatographic peak areas of the component in question in the sample and Standard Solution, respectively.

4.3.2 Ingredient (Bupivacaine Hydrochloride).

4.3.2.1 Infrared Spectrum.

Dissolve about 230 mg in 15 ml of purified water in a separator. In a second separator dissolve about 230 mg of a reference standard in 15 ml of purified water. Treat each solution as follows: Add 1 ml of 10% ammonia solution and extract with three 30 ml portions of chloroform. Evaporate the chloroform at room temperature under a stream of nitrogen. Dry the residue in vacuum. Add 2.0 ml of chloroform, and mix to dissolve.

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Determine the absorption spectra of the prepared solutions of both standard and sample, in 0.1 mm cells between 2 µm and 13 µm with a suitable infrared spectrophotometer, using chloroform in a matched cell as the blank. The spectrum of the solution prepared from the sample shall exhibit maxima only at the same wavelength as that of the reference standard.

A copy of the standard infrared spectrum shall be submitted by the contractor, and shall be kept on file in the Technical & Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center, for use if samples are submitted.

4.3.2.2 Chloride.

Dissolve about 50 mg of sample in 10 ml of purified water in a small separator. Make alkaline with 10% ammonia solution and extract with 10 ml of ether. Drain the aqueous layer into a test-tube and acidify with nitric acid. Add 0.1N silver nitrate. A white, curdy precipitate forms which is insoluble in nitric acid but soluble in a slight excess of 10% ammonia solution.

4.3.2.3 Assay.

Accurately weigh about 600 mg of sample into a 10 ml conical flask. Dissolve in 20 ml of glacial acetic acid and 10 ml mercuric acetate solution (6% in glacial acetic acid). Add 3-5 drops of crystal violet T.S. and titrate to an emerald green end point with 0.1N perchloric acid. Perform a blank determination and make any necessary correction.

1 ml 0.1N $HClO_4$ = 32.49 mg Bupivacaine Hydrochloride (anhydrous)

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4.4 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable paragraphs of PPP-C-PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 50 ml of strength 1, size 1 or strength 2, size 1; or one box containing five ampuls, each ampul containing 50 ml, of strength 1, size 3; or 30 ml of strength 2, size 2 or strength 3, size 2, as specified constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available multiple dose vial for size 1 and a single dose ampul for sizes 2 and 3. The container shall be considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

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5.2.2.2 Unit package.

5.2.2.2.1 Size 1. At the option of the contractor, a unit package may be furnished for each size 1 bottle. If a unit package is furnished, each bottle shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.2.2 Size 2 and size 3. Five ampuls of size 2 or size 3 of one strength only shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Five units of issue of one strength and size only shall be packaged in an intermediate package as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Four intermediate packages (20 units of issue) of one strength and size only of strength 1, size 3; strength 2, size 2; strength 3, size 2; or twenty intermediate packages (100 units of issue) of strength 1, size 1, or strength 2, size 1, shall be packaged in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Four intermediate packages (20 units of issue) of one strength and size only of strength 1, size 3; strength 2, size 2; strength 3, size 2; or twenty intermediate packages (100 units of issue) of strength 1, size 1, or strength 2, size 1 shall be packed in an exterior container as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit packages. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.

(b) the National Stock Number.

NOTES:

1) The National Stock Number shall not be preceded by any prefix.

2) Strip labeling may be used.

3) Label adhesion test shall not apply.

4) If a unit package is not supplied for size 1 bottles, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with 5.5.3 of PPP-C-186. Type I shelf life markings as specified in MIL-STD-129 shall also be shown. Marking shall include the legend:

"DO NOT PERMIT TO FREEZE".

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5.4.3 Exterior container. Exterior container shall be marked in accordance with 5.5.4 of PPP-C-186. Type I shelf life markings as specified in MIL-STD-129 shall also be shown. Marking shall include the legend:

"DO NOT PERMIT TO FREEZE".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification	Specification Designation
6505-00-138-7345	BUPIVACAINE HYDROCHLORIDE INJECTION, 0.25%, 50 ml, 5s	Strength 1, Size 3
6505-00-138-7347	BUPIVACAINE HYDROCHLORIDE INJECTION, 0.50%, 50 ml	Strength 2, Size 1
6505-00-149-3500	BUPIVACAINE HYDROCHLORIDE INJECTION, 0.25%, 50 ml	Strength 1, Size 1
6505-00-149-3501	BUPIVACAINE HYDROCHLORIDE INJECTION 0.50%, 30 ml, 5s	Strength 2, Size 2
6505-00-149-3502	BUPIVACAINE HYDROCHLORIDE INJECTION, 0.75%, 30 ml, 5s	Strength 3, Size 2

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1571

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-B-37245A BUPIVACAINE HYDROCHLORIDE INJECTION

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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